# Darshan Sonagara

Lead Data Scientist · Target · Minneapolis, MN

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Data scientist with about eleven years of professional experience, the last nine in machine learning for search, information retrieval, recommendation, and trust and safety. Lead the Retrieval team for product search at Target; built the company's first semantic search channel and led its launch as a hybrid system serving all Target.com and app search. Previously four years at Goldman Sachs on enterprise search and real-time fraud detection, and graduate research at Rutgers on misinformation detection, cited 331 times.

## EXPERIENCE

### Target · Minneapolis, MN

Apr 2023 – present

**Lead Data Scientist**, Oct 2026 – present

**Senior Data Scientist**, Apr 2023 – Oct 2026

Lead the Retrieval team in Search since April 2024: set the technical direction for product retrieval on Target.com and the Target app, and mentor the team's scientists and engineers.

#### Hybrid product search

Jul 2024 – Feb 2025

Dense retrieval · contrastive learning · PyTorch · AlloyDB · Solr · A/B testing

- Led the design and launch of Target's first hybrid search system, which runs lexical and semantic retrieval in parallel on every search. The work is the basis of a U.S. provisional patent application (first-named inventor) and an arXiv paper (co-first author).
- Built the semantic channel from scratch: selected the embedding model and fine-tuned a dual encoder on about 20 million query–product pairs labeled from purchases, add-to-carts and clicks, using hard negatives and a bidirectional contrastive loss. Field markers and field masking keep it robust to incomplete catalog data.
- Added adaptive precision control, using named-entity recognition and a query classifier to remove off-target vector results (precision@1 from 0.890 to 0.929), and weighted result interleaving, which outperformed reciprocal rank fusion on result lists with little overlap.
- Four-week online A/B tests against lexical-only search: CTR +0.97%, conversion +0.98%, demand per visitor +1.10%, and about 50% fewer zero-result searches. Serving 100% of search traffic since February 2025.

#### Real-time embedding inference service

Feb 2025 – present

Kubernetes · KServe · microservices

- Own the query and item embedding services behind every Target.com and app search, with p99 latency under 50 ms on CPU through caching and adaptive batching.
- The fine-tuned models have become shared infrastructure, used by the Ads, Content Understanding and Query Understanding teams for search ads, item understanding and query normalization.
- Led the services' peak-season readiness with zero incidents, and wrote the runbook other teams adopted as their template.

#### Personalized retrieval

2025

Two-tower models · attention · PyTorch · Vertex AI · PySpark · Redis

- Moved personalization into the retrieval layer with user, query and item towers. Query-guided attention over a shopper's real-time, short-term and long-term behavior produces a user-aware query vector for a dedicated personalized retrieval channel.
- Designed multi-layer cross-attention between the query and each user feature, and an adaptive weighting layer that learns how much purchases, clicks and add-to-carts should each count.
- Built multi-GPU training on Vertex AI, PySpark pipelines for user-history training data and a Redis feature store for near-real-time user embeddings. The best variant improved Recall@K, NDCG and MRR offline.

#### Automated relevance labeling and precision control

2025 – present

Agentic LLM pipelines · LLM fine-tuning · knowledge distillation · cross-encoders

- Designed a three-stage system for four-class search relevance, where each stage covers what the one before it cannot:
  1. **Agentic labeling:** state-of-the-art LLMs in an agentic pipeline produce a first-generation relevance dataset with reasoning for every label. Highest quality, but too slow and costly for all traffic.
  2. **In-house relevance LLM:** fine-tuned on that reasoning data, it labels at scale for evaluation and training at much lower cost, with data kept in-house.
  3. **Online student:** a lightweight cross-encoder or shared bi-encoder, distilled from the in-house LLM, filters off-target results at query time within the search latency budget.

### Goldman Sachs · New York, NY

Feb 2019 – Apr 2023

**Vice President**, Jan 2022 – Apr 2023

**Associate**, Jan 2021 – Dec 2021

**Analyst**, Feb 2019 – Dec 2020

**Summer Intern**, May – Aug 2018

### Enterprise search for compliance surveillance 2022 – Apr 2023

Dense retrieval · result fusion · entity extraction · Elasticsearch

- Led model development for the search platform built for Goldman's Compliance division to find data, conduct surveillance and detect pattern anomalies across petabyte-scale firm records. The platform was later adopted firm-wide.
- Designed and trained in-house retrieval models for high-recall search over sensitive, access-controlled data, where off-the-shelf models could not be used, and the fusion of lexical and semantic results into one ranking. Contributed to the sharding design and to entity extraction that links documents to the firm's knowledge graph.

### Real-time fraud detection for Apple Card 2020 – 2022

Knowledge graphs · graph features · real-time scoring

- Worked on fraud screening for new applications and live transactions on Apple Card, issued by Goldman Sachs since 2019, in a program that grew past 12 million cardholders. Decisions draw on a real-time knowledge graph of related entities and activity.
- Led feature engineering on graph-derived signals and the fraud models built on them, so each application or transaction is judged by its connections and not only its own fields. Contributed to the graph's design and infrastructure.

## Rutgers University · Behavioral Informatics Lab · New Brunswick, NJ Jun 2017 – Dec 2018

**Research Assistant** to Prof. Vivek K. Singh

- Research assistant on NSF-funded cyberbullying detection research (Award IIS-1464287). Designed the multimodal feature set for Instagram cyberbullying detection and led labeling of a new dataset.
- Multimodal fake-news detection: chose the visual and linguistic signals to study, built the pipeline that extracted 43 visual features from news images, and the models that combine them with text features. Published in JASIST (2021), cited 191 times and twice named a Top Cited Article.
- Linguistic fake-news detection: co-authored the SBP-BRiMS 2017 paper and helped build the two credible-news datasets released with it. Cited 98 times.

## Tata Consultancy Services · Kolkata, India Mar 2015 – Dec 2016

**Assistant System Engineer**

- Moved data and CPU-heavy batch jobs from a mainframe system to Hadoop and Hive, reducing mainframe processing cycles, and built a sentiment analyzer for customer product reviews.

## PUBLICATIONS & PATENT

331 citations on Google Scholar (Oct 2026)

**D. Sonagara**, Q. Zhang, A. Singh, A. Li. **Retail Product Search: A Practical Approach at Target.** *arXiv:2609.31498*, 2026. arXiv
Co-first author

V. K. Singh, I. Ghosh, **D. Sonagara**. **Detecting fake news stories via multimodal analysis.** *Journal of the Association for Information Science and Technology*, 72(1): 3–17, 2021. doi:10.1002/asi.24359 191 citations · JASIST Top Cited Article award, 2020–21 and 2021–22

V. K. Singh, R. Dasgupta, **D. Sonagara**, K. Raman, I. Ghosh. **Automated fake news detection using linguistic analysis and machine learning.** *SBP-BRiMS Grand Data Challenge*, Washington, DC, 2017. poster 98 citations · name printed as "D. Sonagra"

**D. Sonagara**, S. Badheka. **Comparison of basic clustering algorithms.** *International Journal of Computer Science and Mobile Computing*, 3(10): 58–61, 2014.

### PATENT APPLICATION

**Hybrid search system with adaptive precision control and weighted result interleaving for product retrieval.** U.S. provisional patent application, filed April 2026, assigned to Target. First-named inventor.

### TALKS & PRESENTATIONS

**Automated Fake News Detection Using Linguistic Analysis and Machine Learning.** Talk and poster, SBP-BRiMS 2017 Grand Data Challenge, George Washington University, Washington, DC, July 2017. Presented by Prof. Vivek K. Singh on behalf of the co-authors. video

## EDUCATION

**M.S. Computer Science**, Rutgers University, New Brunswick, NJ GPA 3.68 / 4.0 2019

**B.E. Information Technology**, G H Patel College of Engineering & Technology, Anand, India GPA 3.70 / 4.0 2015

## SKILLS

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|---|---|
| Search & IR | Hybrid, dense and lexical retrieval; two-tower models; contrastive learning; ANN search; result fusion and interleaving; learning to rank; query understanding; recommender systems |
| LLMs & NLP | LLM fine-tuning and distillation; agentic LLM pipelines; cross-encoders; named-entity recognition and disambiguation |
| Other ML | Knowledge graphs; graph features for fraud detection; multimodal (text and image) models |
| Tools | Python, PyTorch, TensorFlow, Hugging Face, PySpark, Solr, Elasticsearch, AlloyDB, Redis, Kubernetes, KServe, Vertex AI, GCP |